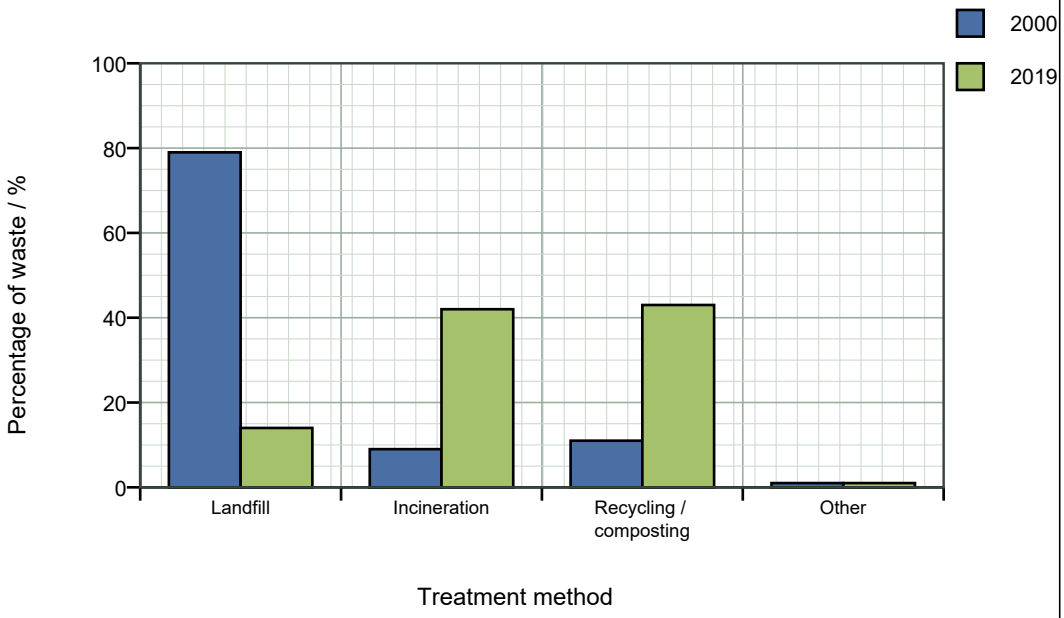
Questions by topic: Solid wastes - Set 1 (55 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1



0 1 . 1 Describe the changes shown in Figure 1.

[2 marks]

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0 1 . 2 Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

1 .....

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2 .....

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3 .....

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0 1 . 3 Describe **four** features of a well-managed landfill site.

[4 marks]

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1 .....

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2 .....

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4 .....

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0 1 . 4 Suggest why incineration has increased more than recycling for some waste types.

[1 mark]

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0 2 . 1 Give **three** advantages of incinerating domestic waste.

[3 marks]

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2 .....

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3 .....

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0 2 . 2 Explain how each of the following features of a modern incinerator reduces pollution.

[4 marks]

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High combustion temperature with rapid cooling of gases .....

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Water spray scrubber .....

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Removal of wet / compostable waste .....

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Removal of recyclable waste .....

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0 2 . 3 Explain why the ash from waste incinerators needs careful disposal.

[2 marks]

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0 2 . 4 State **one** reason why incineration is expensive.

[1 mark]

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0 3 Table 1 shows three specialist solid wastes.

0 3 . 1 Complete Table 1 to show the main hazard of each waste and the disposal method used.

[6 marks]

Table 1

Waste	Main hazard	Disposal method	
Asbestos			
Cyanide compounds			
High-level radioactive waste			

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0 3 . 2 Explain why encapsulated asbestos in cement roof panels is not considered dangerous while it is intact.

[1 mark]

0 3 . 3 Describe the process of encapsulation used for intermediate-level radioactive waste and wastes containing heavy metals.

[2 marks]

0 3 . 4 Explain why vitrification is preferred to encapsulation in cement for high-level waste.

[1 mark]

10

0 4 A coal-mine spoil heap on a hillside has been abandoned. The spoil contains shale with some iron pyrites and residual coal.

0 4 . 1 Explain **three** problems this spoil heap could cause and describe how each could be managed.

[6 marks]

0 4 . 2 Explain why deep layers of soil are applied when a spoil heap containing toxic materials is restored.

[2 marks]

0 4 . 3 Explain why an old steel-works spoil heap may be valuable for wildlife conservation.

[2 marks]

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0 5 . 1 Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

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0 5 . 2 Give **three** factors that affect the choice of disposal method for a solid waste.

[3 marks]

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2 .....

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3 .....

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0 5 . 3 Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill.

[2 marks]

Advantage .....

.....

Disadvantage .....

.....

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0 5 . 4 Explain why mixed municipal waste is difficult to recycle.

[1 mark]

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0 6 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 6 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 6 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

1 .....

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2 .....

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5

END OF QUESTIONS

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